

Problem-Solution Fit

Date	15 March 2026
Team ID	SWTID-2026-6771
Project Name	Credit card Approval Prediction
Maximum Marks	5 Marks

1. CUSTOMER SEGMENT(S) [CS] • Individuals applying for a credit card • Banks and financial institutions • Loan/Credit officers	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited financial knowledge • Lack of credit history • Time constraints • Internet/device availability	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual Verification ✓ Human judgment ✗ Slow and error-prone Traditional Rule-Based Systems ✓ Simple to implement ✗ Less accurate and inflexible
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Manual approval process is slow (High) • Inconsistent decision-making (Medium) • Long waiting time for applicants (High)	9. PROBLEM ROOT / CAUSE [RC] • Large number of applications • Time-consuming manual verification • Human bias and inconsistency • Difficulty analyzing multiple applicant factors simultaneously	7. BEHAVIOR + ITS INTENSITY [BE] • Frequently checks eligibility before applying (High) • Wants quick decisions (High) • Compares different banks (Medium)
3. TRIGGERS TO ACT [TR] • Customer applies for a credit card • Need for instant eligibility check • Bank wants to automate	10. YOUR SOLUTION [SL] Develop a Machine Learning-based Credit Card Approval Prediction System using Random	8. CHANNELS OF BEHAVIOR [CH] ONLINE Bank websites, Mobile apps, Credit card portals

approvals

Forest and **Flask** to analyze applicant details and predict approval/rejection instantly. The system provides faster, consistent, and data-driven decisions through an easy-to-use web interface.

4. EMOTIONS

BEFORE / AFTER [EM]

Before: Anxious, uncertain, hopeful

After: Happy if approved; disappointed but informed if rejected

OFFLINE

Bank branches, Customer service representatives